

Trajectory Optimisation for a Steerable Airdrop in Unknown Wind

Guiding packages to a target for relief distribution,
with no wind sensor on board.

Amulya D · Atomic Energy Central School, Mysuru

80,000

UNREACHED · ASSAM FLOODS · JULY 2026

23.9 m

HOW FAR AN UNGUIDED DROP LANDS FROM THE TARGET



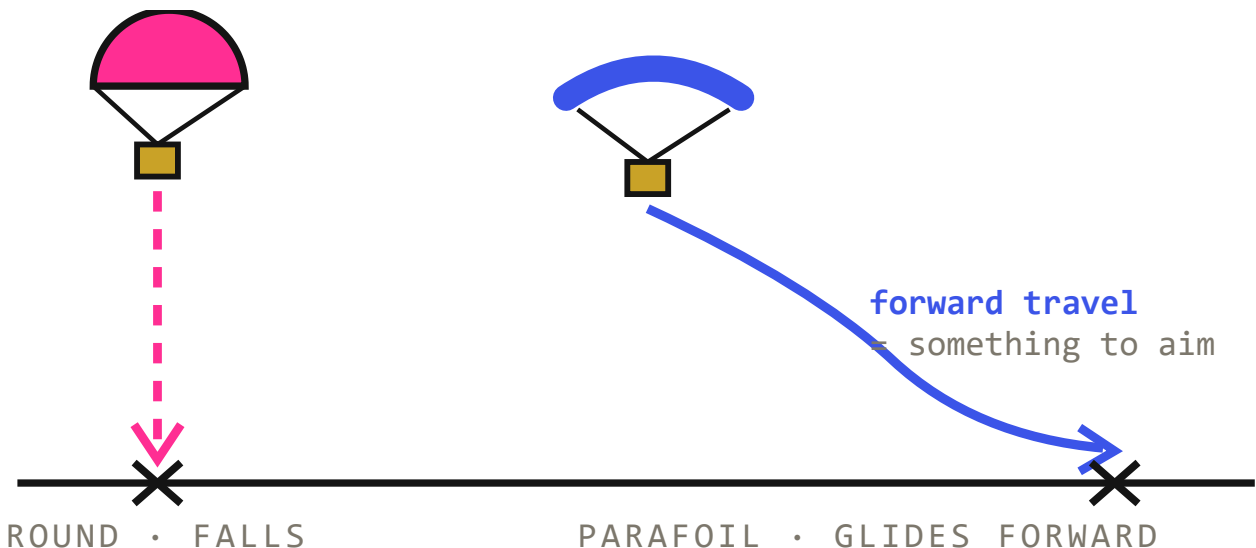
SCAN 1 OF 2

Watch it miss.

In your own room. Point your phone at the floor.

ANSWER: YES, IF IT IS A WING

You can steer it. You cannot stop it.



$$v = \sqrt{(2mg / \rho S C_L)}$$

Only the wing loading m/S appears.

No engine, no brakes, constant sink rate. **Time in the air is fixed at release** — so the distance it will fly is too.

WHAT IT WILL FLY BEFORE IT LANDS



DISTANCE TO THE TARGET



It has to spend the surplus.

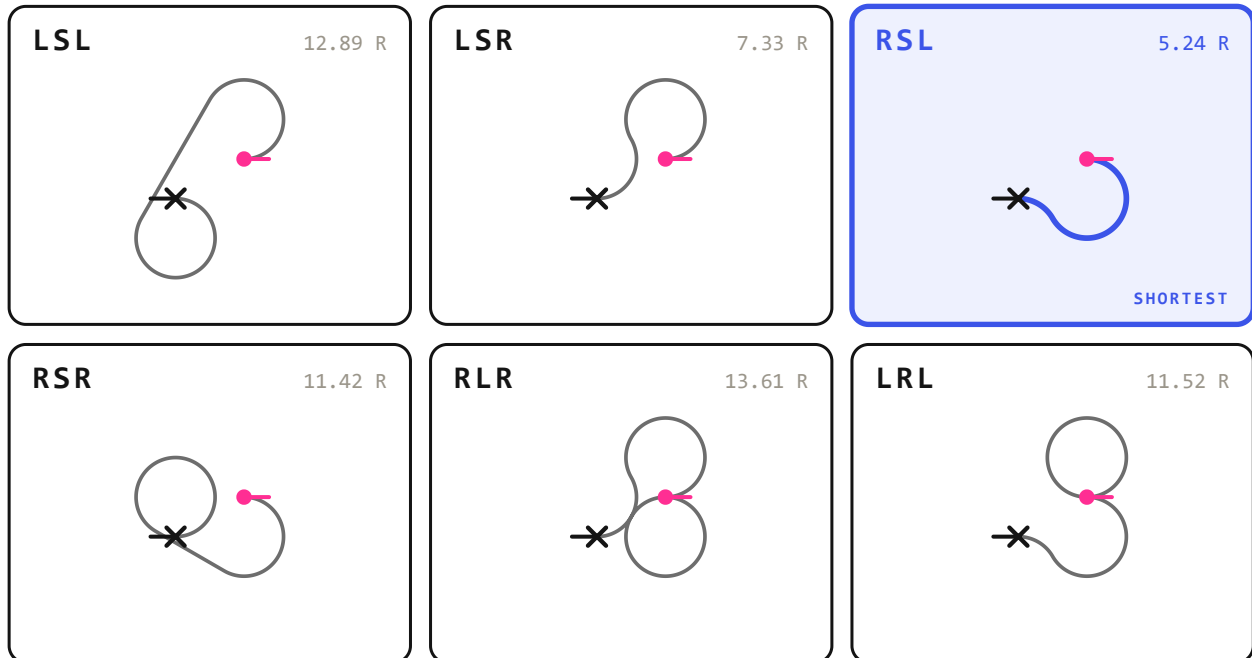
SO → Fly straight at the target and it arrives with height to spare.

So it weaves — on purpose.

$$\cos \alpha = d / s$$

d = distance to go · s = flying left. Stretches the path until the budget is gone.

And d is a curve, not a line — it cannot turn on the spot. **Dubins, 1957: the shortest such curve is one of six.**



Same start, target and headings, solved six ways at one scale.
Computed by the model, not drawn.

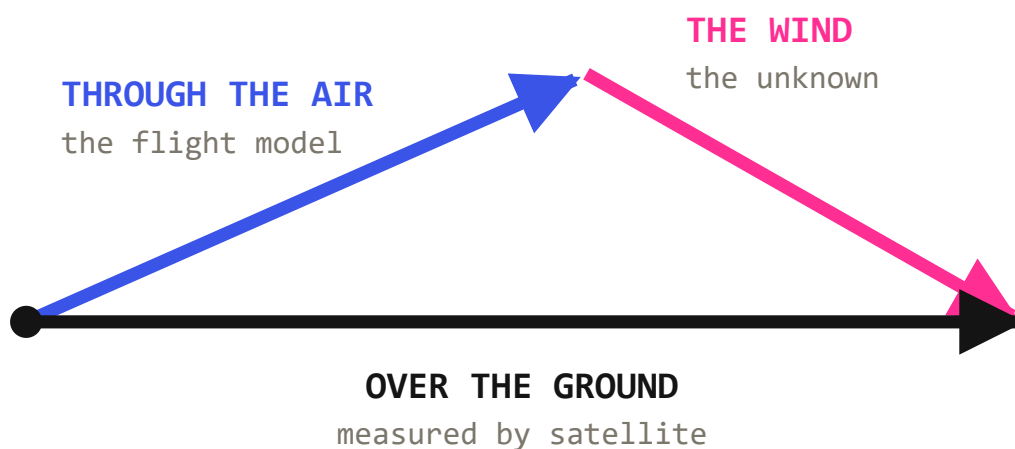
Dubins measures; the weave spends.

S0 → It is turning most of the way down.

THE PART THAT PAYS FOR ITSELF

And that is how it finds the wind.

No wind sensor. The wind is a **subtraction**:



Except it should be impossible

FLYING STRAIGHT

Headwind, or too slow? **Same track.**

TURNING

Different tracks. Solvable.

And by panel 03 it is *already turning*. **The measurement is free.**

0.104 m s⁻¹

MEDIAN WIND ERROR RECOVERED

BUT → **Does it land closer?**

150 RANDOMISED DROPS

Yes — less than half the miss.

WHAT IT KNOWS

MEDIAN MISS

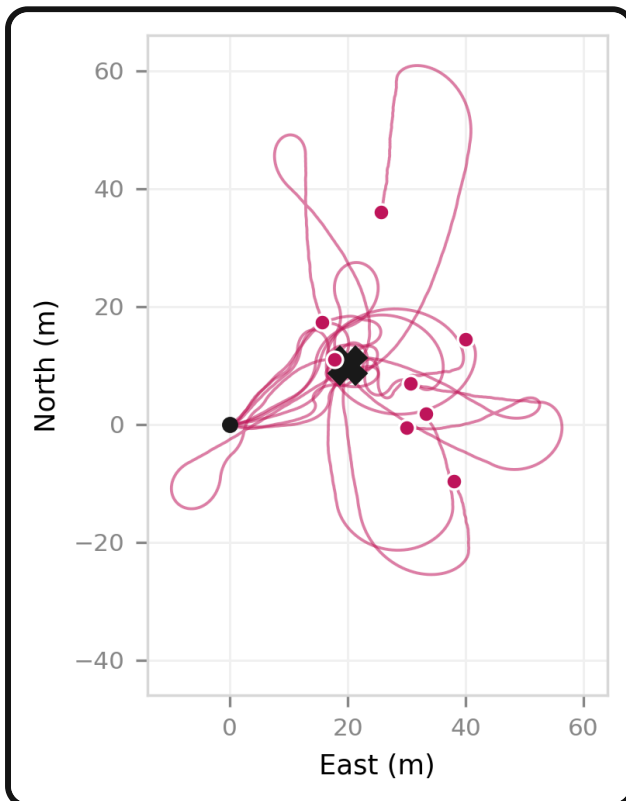
Nothing

8.57 m

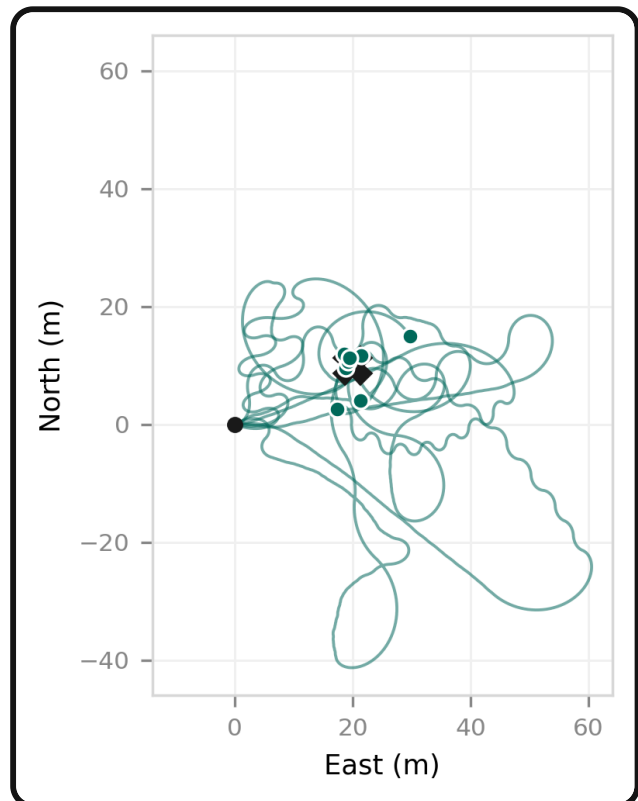
Works it out**3.89 m**

Told the truth

4.62 m



No estimate - scattered.



EKF - on target.

Closer in 112 of 150 drops.

What it does not show. The EKF and true-wind intervals overlap, so at this sample size those two are not separated.

SO → Build the most capable canopy you can afford?

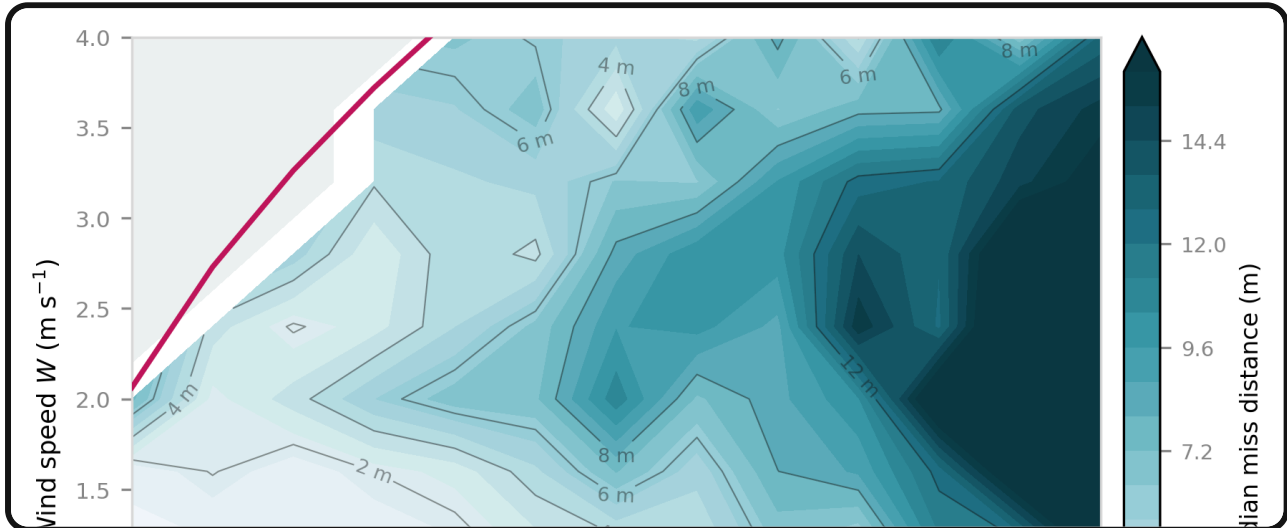
THE ONE NUMBER THAT DECIDES EVERYTHING

No. A heavier one is worse.

Beating wind W forces loading up as W^2 . But turn radius rises with loading — and that is the smallest correction left at the end.

$$R_{\min} \propto W^2$$

The error floor follows the wind you designed for, not the wind you meet.



Loading against wind speed, 24 drops each. **The dark valley moves with the wind.**

Loading is set by **ballast** — matched to the forecast on the morning of the drop.



SCAN 2 OF 2

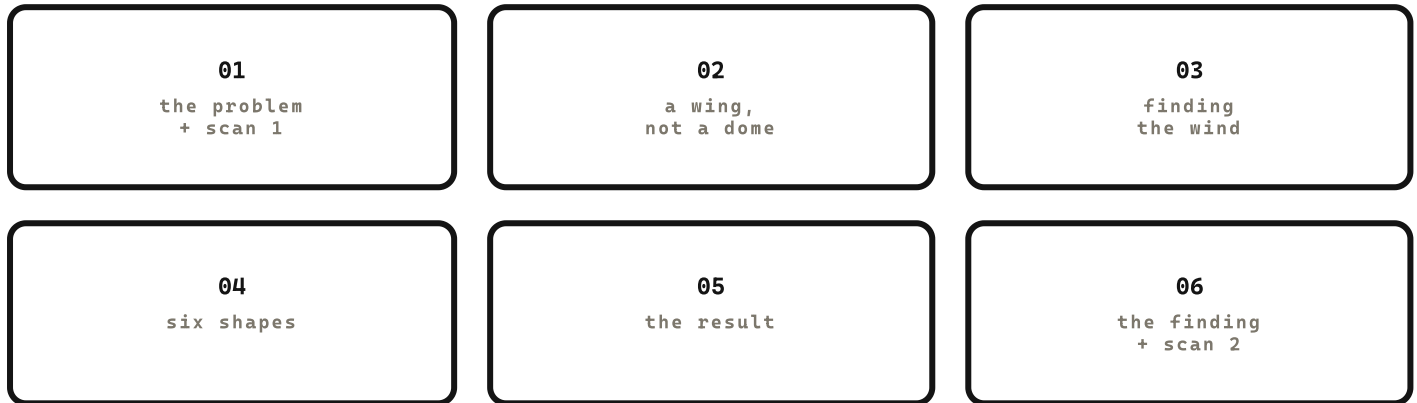
Now watch it land.

Same drop, same wind. Dotted line is where it went before.

✓ **23.9 m off → 2.35 m, at your feet.**

Sticking it together

Six sheets, three across and two down. Each panel is complete in itself, so **nothing needs trimming and nothing crosses a join**. The white border of each sheet becomes the gutter between panels.



1. Print pages 1–6 on A4 at **“Actual size” / 100%**, so the panels come out the size the layout was built for and the margins match. (The codes themselves were tested down to 70% and still scan — it is the layout that wants full size, not them.)
2. Lay all six out on the backing sheet *before* gluing anything. Line up the tops of panels 01–03, then 04–06.
3. Glue the middle sheet of the top row first and work outwards, so any error is shared between both sides instead of accumulating at one edge.
4. **Scan both QR codes with your own phone once the poster is up.** Test under the lighting in the hall, not at home.

If a QR code will not scan

Both codes were decoded from this very PDF at every resolution from draft to 300 dpi, so the artwork is not the problem. In practice it is nearly always **glare** — a reflection off tape, glue or a glossy sheet, sitting across the middle of the code. Move so the light is behind you rather than behind the phone. Failing that, low ink leaves gaps in the modules: reprint that one sheet. The codes carry the highest error correction level, so a crease or a fingerprint will not stop them.

Reading order

01 sets up the problem and sends them to the first scan. 02–04 are the model: why it can steer, how it finds the wind, how far it really has to fly. 05 is the evidence, including what it does *not* show. 06 is the finding, and sends them to the second scan — so a judge who only reads the top row still gets the problem, the mechanism and a demonstration.

Full paper and the interactive model: **1101-hub.github.io/guided-airdrop**